

## Problem 2: Solving the KKT system (0.5 pt)

Solve the following convex quadratic problem in  $\mathbb{R}^3$  with linear equality constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) = (-3 \ -2 \ 7)x + \frac{1}{2}x^\top \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix} x \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 2 \end{pmatrix}. \end{aligned}$$

*Hint:* use Gaussian elimination or QR decomposition to solve the resulting KKT system.

**Solution 2.** We write the problem in the standard quadratic form

$$\min_{x \in \mathbb{R}^3} \frac{1}{2}x^\top Qx + c^\top x \quad \text{s.t.} \quad Cx = d,$$

with

$$Q = \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix}, \quad c = \begin{pmatrix} -3 \\ -2 \\ 7 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \quad d = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Introducing Lagrange multipliers  $\lambda \in \mathbb{R}^2$  for the equality constraints, the KKT conditions are

$$\begin{aligned} Qx + c + C^\top \lambda &= 0, \\ Cx - d &= 0. \end{aligned}$$

Equivalently, we obtain the linear KKT system

$$\begin{pmatrix} 6 & 0 & 2 & 1 & 0 \\ 0 & 4 & 0 & 1 & 1 \\ 2 & 0 & 6 & 2 & 1 \\ 1 & 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -7 \\ 0 \\ 2 \end{pmatrix},$$

where the right-hand side comes from  $Qx + C^\top \lambda = -c$  and  $Cx = d$ .

We now solve this linear system using Gaussian elimination. The augmented matrix is

$$\left[ \begin{array}{ccccc|c} 6 & 0 & 2 & 1 & 0 & 3 \\ 0 & 4 & 0 & 1 & 1 & 2 \\ 2 & 0 & 6 & 2 & 1 & -7 \\ 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 2 \end{array} \right].$$

Applying Gaussian elimination (row operations), we reduce it to row-echelon form and then to reduced row-echelon form. The resulting matrix is

$$\left[ \begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 11 \\ 0 & 0 & 0 & 0 & 1 & -21 \end{array} \right],$$

which is the reduced row-echelon form of the KKT system.

From this we read off the unique solution

$$x_1^* = -1, \quad x_2^* = 3, \quad x_3^* = -1,$$

and the corresponding Lagrange multipliers

$$\lambda_1^* = 11, \quad \lambda_2^* = -21.$$

One can verify that the constraints are satisfied:

$$\begin{aligned} x_1^* + x_2^* + 2x_3^* &= -1 + 3 + 2(-1) = 0, \\ x_2^* + x_3^* &= 3 + (-1) = 2. \end{aligned}$$

Since  $Q$  is positive definite, the quadratic objective is strictly convex and the KKT conditions are sufficient for optimality. Therefore  $x^* = (-1, 3, -1)^\top$  is the unique minimizer of the problem.

The minimal objective value is

$$f(x^*) = \begin{pmatrix} -3 & -2 & 7 \end{pmatrix} x^* + \frac{1}{2} (x^*)^\top Q x^* = 16.$$